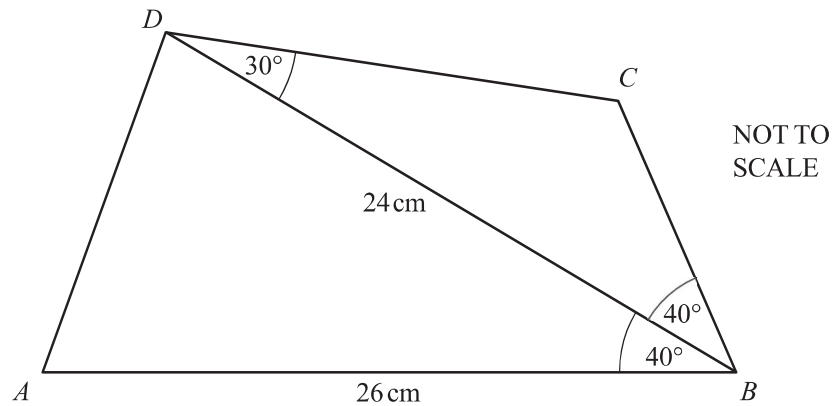


Question 1



$ABCD$ is a quadrilateral and BD is a diagonal.
 $AB = 26\text{ cm}$, $BD = 24\text{ cm}$, angle $ABD = 40^\circ$, angle $CBD = 40^\circ$ and angle $CDB = 30^\circ$.

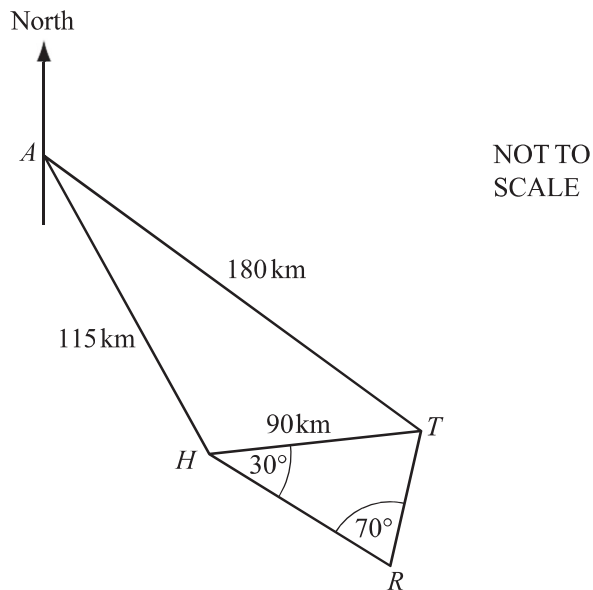
(a) Calculate the area of triangle ABD . [2]

(b) Calculate the length of AD . [4]

(c) Calculate the length of BC . [4]

(d) Calculate the shortest distance from the point C to the line BD . [2]

Question 2



The diagram shows some straight line distances between Auckland (A), Hamilton (H), Tauranga (T) and Rotorua (R).

$AT = 180$ km, $AH = 115$ km and $HT = 90$ km.

(a) Calculate angle HAT .

Show that this rounds to 25.0° , correct to 3 significant figures.

[4]

(b) The bearing of H from A is 150° .

Find the bearing of

(i) T from A ,

[1]

(ii) A from T .

[1]

(c) Calculate how far T is east of A . [3]

(d) Angle $THR = 30^\circ$ and angle $HRT = 70^\circ$.

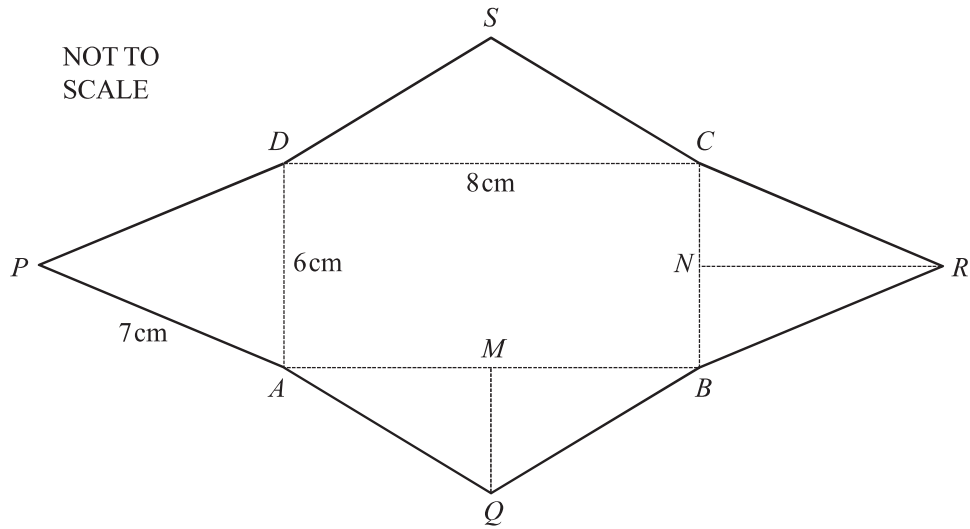
Calculate the distance TR . [3]

(e) On a map the distance representing HT is 4.5cm.

The scale of the map is $1 : n$.

Calculate the value of n . [2]

Question 3



The diagram above shows the net of a pyramid.

The base $ABCD$ is a rectangle 8 cm by 6 cm .

All the sloping edges of the pyramid are of length 7 cm .

M is the mid-point of AB and N is the mid-point of BC .

(a) Calculate the length of

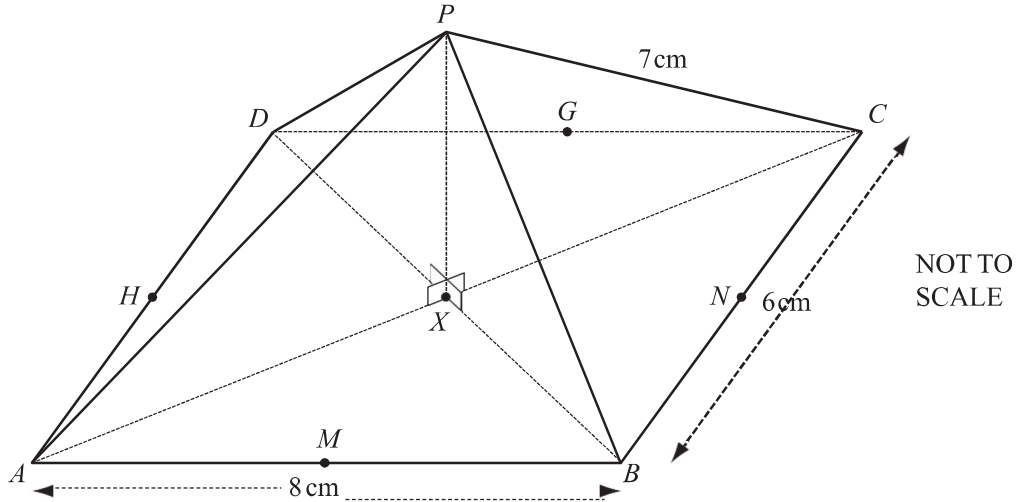
(i) QM , [2]

(ii) RN . [1]

(b) Calculate the surface area of the pyramid. [2]

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(c)



The net is made into a pyramid, with P , Q , R and S meeting at P .

The mid-point of CD is G and the mid-point of DA is H .

The diagonals of the rectangle $ABCD$ meet at X .

- (i) Show that the height, PX , of the pyramid is 4.90 cm, correct to 2 decimal places. [2]

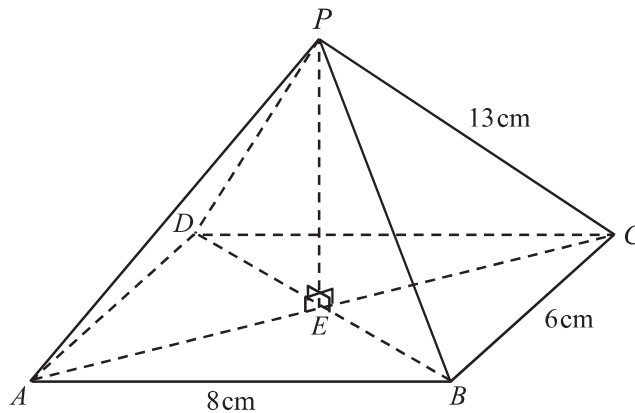
- (ii) Calculate angle PNX . [2]

- (iii) Calculate angle HPN .

- (iv) Calculate the angle between the edge PA and the base $ABCD$. [3]

- (v) Write down the vertices of a triangle which is a plane of symmetry of the pyramid. [1]

Question 4



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The diagram shows a pyramid on a horizontal rectangular base $ABCD$.

The diagonals of $ABCD$ meet at E .

P is vertically above E .

$AB = 8$ cm, $BC = 6$ cm and $PC = 13$ cm.

(a) Calculate PE , the height of the pyramid. [3]

(b) Calculate the volume of the pyramid.

[The volume of a pyramid is given by $\frac{1}{3} \times \text{area of base} \times \text{height}.$] [2]

(c) Calculate angle PCA . [2]

(d) M is the mid-point of AD and N is the mid-point of BC .
Calculate angle MPN .

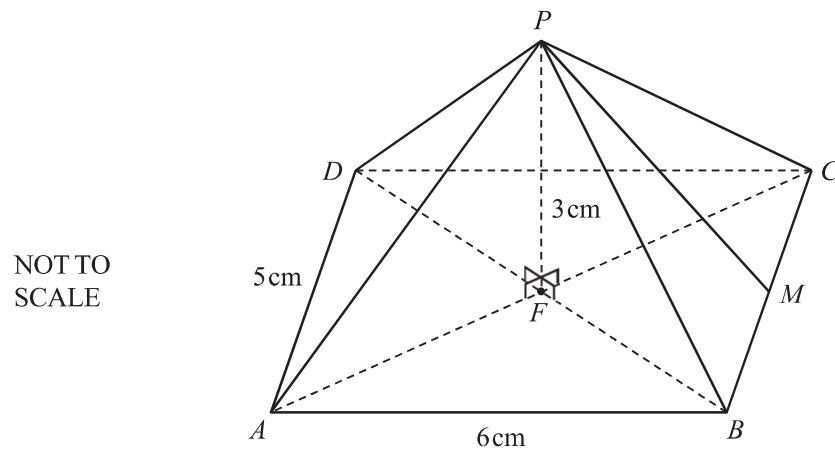
[3]

(e) (i) Calculate angle PBC . [2]

(ii) K lies on PB so that $BK = 4$ cm.
Calculate the length of KC .

[3]

Question 5



The diagram shows a pyramid on a rectangular base $ABCD$, with $AB = 6\text{ cm}$ and $AD = 5\text{ cm}$.
The diagonals AC and BD intersect at F .
The vertical height $FP = 3\text{ cm}$.

(a) How many planes of symmetry does the pyramid have? [1]

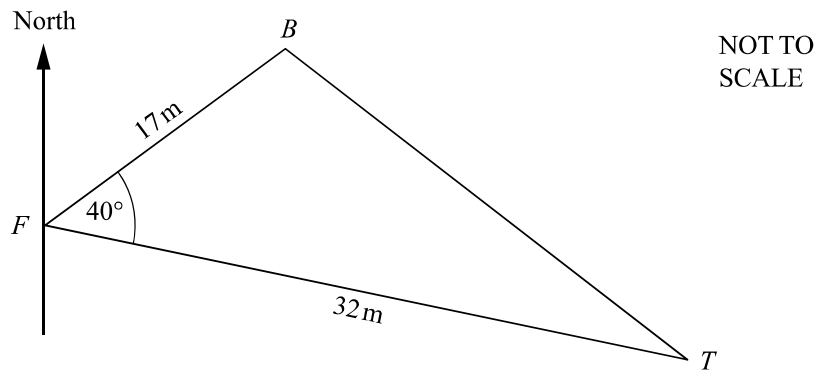
(b) Calculate the volume of the pyramid.
[The volume of a pyramid is $\frac{1}{3} \times \text{area of base} \times \text{height}$.] [2]

(c) The mid-point of BC is M .
Calculate the angle between PM and the base. [2]

(d) Calculate the angle between PB and the base. [4]

(e) Calculate the length of PB . [2]

Question 6



Felipe (F) stands 17 metres from a bridge (B) and 32 metres from a tree (T). The points F , B and T are on level ground and angle $BFT = 40^\circ$.

(a) Calculate

(i) the distance BT ,

[4]

[3]

(ii) the angle BTf .

(b) The bearing of B from F is 085° . Find the bearing of

(i) T from F ,

[1]

(ii) F from T ,

[1]

(iii) B from T .

[1]

(c) The top of the tree is 30 metres vertically above T . Calculate the angle of elevation of the top of the tree from F .

[2]